

APPENDIX

Name	Symbol	Value	Comment
System Parameterization			
Discharge coefficient	c_d	0.6	Typical value for water [1]
Tank base area	A	$\pi \cdot 1 \text{ m}^2$	Cyl. tank with $r = 1 \text{ m}$
Opening area	A_o	$\pi \cdot 0.01 \text{ m}^2$	Opening with $r = 0.1 \text{ m}$
Gravitational acceleration	g	$9.81 \text{ m}^2 \text{ s}^{-1}$	
Initial state	$\mathbf{x}_0$	1.5 m	
State space	$\mathcal{X}$	[0 m, 3 m]	Normalized to -1 and 1 in Implementation
Input space	$\mathcal{U}$	$[0 \text{ m}^3 \text{ s}^{-1}, 0.2 \text{ m}^3 \text{ s}^{-1}]$	Normalized to -1 and 1 in implementation
Sampling time	τ	5 s	
ODE solver		Tsit5	Proposed by [2], DiffraX implementation [3]
Excitation Experiments Parameterization			
Prediction horizon	L	10	
Number of grid points	M	2500	Uniformly spaced, 50 per dimension
Input optimizer step size	α	0.1	
Optimizer steps per iteration	I	10	
Penalty factor states	λ_x	1	
Penalty factor inputs	λ_u	1	
KDE bandwidth	h	≈ 0.0365	
Penalty order		2	
Clip actions		Enabled	Force actions to be within $\mathcal{X}$
Parallel starts		5	Number of parallel starts in multistart opt.
Warm start		Enabled	Reuse last optimization result in one start
Model optimizer step size	β	10^{-4}	
Model batch size		128	
Train steps per iteration	J	1	
Model horizon	R	10	
Model depth		3	
Model width		128	
Featurization		Identity	The data is used as is for the model
Start learning at		10	How many steps are done before learning starts
Seeds		[101, 130]	Initial SE experiment seeds
Modeling Experiments Parameterization			
Model optimizer step size		10^{-4}	
Model batch size		128	
Train steps		100 000	
Model horizon		2	
Model depth		2	
Model width		64	
Featurization		Identity	The data is used as is for the model
Seeds		[0, 9]	Initial system modeling experiment seeds
Safe Set Approximation Parameterization			
Number of state grid elements	$M_{r,ci}^{(\mathbf{x})}$	50	
Number of action grid elements	$M_{r,ci}^{(\mathbf{u})}$	50	
Number of grid elements	$M_{r,ci}^{(\mathbf{x},\mathbf{u})}$	2500	
Reachability horizon	N_r	200	
Reachability parallel starts		100	Number of parallel starts in multistart opt.
Reachability optimization steps		50 000	
Reachability tolerance value	ϵ_r	10^{-4}	
Control invariant set horizon	N_{ci}	200	
Control invariant set parallel starts		10	

TABLE I: Fluid tank parameters. The tank is assumed to be a cylindrical tank.

Name	Symbol	Value	Comment
System Parameterization			
Pendulum mass	m	1 kg	
Pendulum length	l	1 m	
Gravitational acceleration	g	$9.81 \text{ m}^2 \text{ s}^{-1}$	
Initial state	$\mathbf{x}_0$	$[0 \ 0 \text{ s}^{-1}]$	
State space	$\mathcal{X}$	$[-\pi, \pi] \times [-10 \text{ s}^{-1}, 10 \text{ s}^{-1}]$	Normalized to -1 and 1 in implementation
Input space	$\mathcal{U}$	$[-5 \text{ N m}, 5 \text{ N m}]$	Normalized to -1 and 1 in implementation
Sampling time	τ	0.02 s	
ODE solver		Tsit5	Proposed by [2], Diffrax implementation [3]
Excitation Experiments Parameterization			
Prediction horizon	L	20	
Number of grid points	M	9261	Uniformly spaced, 21 per dimension
Input optimizer step size	α	0.1	
Optimizer steps per iteration	I	10	
Penalty factor states	λ_x	1	
Penalty factor inputs	λ_u	1	
KDE bandwidth	h	≈ 0.1063	
Penalty order		2	
Clip actions		Enabled	Force actions to be within $\mathcal{X}$
Parallel starts		5	Number of parallel starts in multistart opt.
Warm start		Enabled	Reuse last optimization result in one start
Model optimizer step size	β	10^{-4}	
Model batch size		128	
Train steps per iteration	J	1	
Model horizon	R	20	
Model depth		3	
Model width		128	
Featurization		Wrap theta	Use sin and cos of angle
Start learning at		20	How many steps are done before learning starts
Seeds		[101, 130]	Pseudo random number generation seeds
Modeling Experiments Parameterization			
Model optimizer step size		10^{-4}	
Model batch size		128	
Train steps		100 000	
Model horizon		2	
Model depth		2	
Model width		64	
Featurization		Wrap theta	Use sin and cos of angle
Seeds		[0, 9]	Initial system modeling experiment seeds
Safe Set Approximation Parameterization			
Number of state grid elements	$M_{\text{r,ci}}^{(\mathbf{x})}$	2500	
Number of action grid elements	$M_{\text{r,ci}}^{(\mathbf{u})}$	50	
Number of grid elements	$M_{\text{r,ci}}^{(\mathbf{x}, \mathbf{u})}$	125 000	
Reachability horizon	N_r	200	
Reachability parallel starts		400	Number of parallel starts in multistart opt.
Reachability optimization steps		50 000	
Reachability tolerance value	ϵ_r	10^{-4}	

Control invariant set horizon	N_{ci}	200
Control invariant set parallel starts		10
Control invariant set optimization steps		50 000

TABLE II: Pendulum parameters.

Name	Symbol	Value	Comment
System Parameterization			
Pendulum mass	m_p	0.1 kg	
Cart mass	m_c	1 kg	
Pendulum length	l	0.5 m	
Friction cart-track	μ_c	0.5	
Friction pendulum-cart	μ_p	0.002	
Gravitational acceleration	g	$9.81 \text{ m}^2 \text{ s}^{-1}$	
Initial state	$\mathbf{x}_0$	$[0 \text{ m} \quad 0 \text{ m s}^{-1} \quad 0 \quad 0 \text{ s}^{-1}]$	
State space	$\mathcal{X}$	$[-2.4 \text{ m}, 2.4 \text{ m}]$ $\times [-8 \text{ m s}^{-1}, 8 \text{ m s}^{-1}]$ $\times [-\pi, \pi]$ $\times [-8 \text{ s}^{-1}, 8 \text{ s}^{-1}]$	Normalized to -1 and 1 in implementation
Input space	$\mathcal{U}$	$[-10 \text{ N}, 10 \text{ N}]$	Normalized to -1 and 1 in implementation
Sampling time	τ	0.02 s	
ODE solver		Tsit5	Proposed by [2], DiffraX implementation [3]
Excitation Experiments Parameterization			
Prediction horizon	L	50	
Number of grid points	R	10 000	Uniformly spaced, 10 per dimension
Input optimizer step size	α	0.1	
Optimizer steps per iteration	I	5	
Penalty factor states	λ_x	1	
Penalty factor inputs	λ_u	1	
KDE bandwidth	h	0.12	
Penalty order		2	
Clip actions		Enabled	Force actions to be within $\mathcal{X}$
Parallel starts		5	Number of parallel starts in multistart opt.
Warm start		Enabled	Reuse last optimization result in one start
Model optimizer step size	β	10^{-4}	
Model batch size		128	
Train steps per iteration	J	10	
Model horizon	R	50	
Model depth		3	
Model width		128	
Featurization		Wrap theta	Use sin and cos of angle
Start learning at		50	How many steps are done before learning starts
Seeds		[101, 130]	Pseudo random number generation seeds
Modeling Experiments Parameterization			
Model optimizer step size		10^{-4}	
Model batch size		128	
Train steps		100 000	
Model horizon		2	
Model depth		2	
Model width		64	
Featurization		Wrap theta	Use sin and cos of angle
Seeds		[0, 9]	Initial system modeling experiment seeds

Safe Set Approximation Parameterization		
Number of state grid elements	$M_{r,ci}^{(\mathbf{x})}$	390 625
Number of action grid elements	$M_{r,ci}^{(\mathbf{u})}$	25
Number of grid elements	$M_{r,ci}^{(\mathbf{x},\mathbf{u})}$	9 765 625
Reachability horizon	N_r	200
Reachability parallel starts		10
Reachability optimization steps		50 000
Reachability tolerance value	ϵ_r	10^{-4}
Control invariant set horizon	N_{ci}	200
Control invariant set parallel starts		10
Control invariant set optimization steps		50 000

Number of parallel starts in multistart opt.

TABLE III: Cart pole parameters.

REFERENCES

- [1] W. Palm III, *System dynamics*. McGraw-Hill Higher Education, 2010.
- [2] C. Tsitouras, “Runge–kutta pairs of order 5 (4) satisfying only the first column simplifying assumption,” *Computers & Mathematics with Applications*, vol. 62, no. 2, pp. 770–775, 2011.
- [3] P. Kidger, *On Neural Differential Equations*. PhD thesis, University of Oxford, 2021.